

Abstract

This paper builds on linguistic case theory and cognitive science and provides a unified algebraic framework for cognitive systems modeling centered on the *case diagram*. Formally, case systems are modeled as categories whose objects are case roles and whose morphisms are grammatical relations; alignment typologies (nominative-accusative, ergative-absolutive, active-stative, tripartite, and fluid-S) are captured as structure-preserving functors between these categories. Within this framework, sentence structure becomes a compact-closed string diagram—computable via DisCoCat and, at the discourse level, DisCoCirc—that functions not merely as descriptive notation but as an executable witness of grammaticality grounded in situated reasoning. Historically, the analysis traces from Pāṇini's *kāraṇa* theory through Jakobson's feature matrices and Fillmore's deep cases to Dowty's graded proto-roles, showing how morphosyntactic inflection supplies the structural skeleton for relational differentiation in natural and designed languages.

Equipping case categories with proto-role satisfaction yields